HOSTILE PRE-PILOT INTEGRITY AUDIT | METHODOLOGICAL & DATA-INTEGRITY REVIEW | PHASE 4.1

SimuPharm Phase 4.1
Hostile Pre-Pilot Integrity Audit

Educational Validation Study — Recruitment Readiness Assessment

AUDIT DATE: October 4, 2026	**DOCUMENT ID:** AUDIT-P4.1-HOSTILE-PREPILOT-2026-REV1	**EVALUATION PACKAGE:** Phase 4 Pilot Package	**CODEBASE:** FROZEN (Zero Modifications)

AUDITOR GOVERNANCE DECLARATION

Mandate: This audit was conducted under an explicit hostile charter to determine whether the Phase 4 educational pilot package is methodologically sound, internally consistent, psychometrically valid, ethically compliant, and safe to execute *before any real student participant is recruited*.

Code Freeze & Non-Fabrication Certification: Zero source code modifications were performed on the SimuPharm engine (`hplc_simulator_script.js`, `script.js`, `index.html`). Zero participant data, fake p-values, simulated effect sizes, or artificial SUS scores were generated. All educational outcomes remain strictly unobserved.

2. EXECUTIVE SUMMARY

An exhaustive adversarial audit was conducted across all 44 artifacts comprising the Phase 4 educational research design, psychometric instruments, ethics declarations, data schemas, analysis pipelines, and manuscript outlines.

Overall Methodological Status: The Phase 4 package demonstrates outstanding structural organization, reproducible client-side software architecture, complete separation of testing data, and rigorous adherence to the Phase 3.5 claim boundaries. However, the package contains **three Critical Blockers**, **five Major Methodological Issues**, and **four Minor Inconsistencies** that compromise data validity and would lead to severe peer-review rejection if uncorrected prior to participant recruitment.

Audit Dimension	Blockers	Major	Minor	Readiness Status	Primary Methodological Finding
Ethics & Institutional Approval	1	0	0	NO-GO (PENDING)	Self-declaration of 45 CFR 46.104(d)(1) exemption is legally insufficient; formal institutional review/authorization is mandatory.
Data Architecture & Schema	1	1	1	CRITICAL BLOCKER	12-point practical task rubric cannot be entered into `participant_data_template.csv` due to missing columns for reasoning and actions.
Assessment & Psychometrics	1	2	1	CRITICAL BLOCKER	Instructor script in Phase 3 spoon-feeds exact post-test answers; 80% of post-test items are keyed to option "B" (position bias).
Statistical Analysis Plan (SAP)	0	2	1	MAJOR REVISIONS	Normality-contingent test selection distorts Type I error; Wilcoxon signed-rank script drops zero-gain tied scores.
Usability & SUS Instrument	0	1	0	MAJOR REVISIONS	Modified wording in SUS items 1 and 7 invalidates comparison against standard Brooke (1996) / Bangor benchmark databases.
Manuscript & Claims	0	0	1	SOUND	Phase 3.5 claim boundaries preserved; explicit placeholder tokens (`[N]`, `[MEAN]`) maintained.

PROVISIONAL AUDIT DECISION: NO-GO — MAJOR METHODOLOGICAL CORRECTIONS REQUIRED

Participant recruitment must remain **STRICTLY HALTED**. While the educational concept is innovative, launching the pilot under the current protocol would result in fatal data collection omissions (inability to store practical reasoning data), massive instructor cueing bias, psychometric response-position distortion, and regulatory non-compliance.

3. SOURCE ARTIFACT INVENTORY

All 44 artifacts located in `SIMUPHARM_PHASE4_EDUCATIONAL_VALIDATION/` were audited for structural integrity, internal consistency, and absence of fabricated participant data.

Folder / Module	Artifact Filename	Size	Role	Audit Verification Status
`00_PHASE4_README/`	`PHASE4_SCOPE.md`	4.6 KB	Scope & Boundaries	**CLEAN** Disavows AI/ML, digital twins, and 2.07% general accuracy.
`00_PHASE4_README/`	`PHASE4_DECISION_STATUS.md`	2.5 KB	Gate Register	**CLEAN** Correctly leaves institutional approval box unchecked.
`00_PHASE4_README/`	`ARTIFACT_INDEX.md`	5.6 KB	Master Index	**CLEAN** All files indexed; zero data contamination.
`01_RESEARCH_DESIGN/`	`research_questions.md`	3.1 KB	Hypotheses	**CLEAN** Non-directional primary hypothesis.
`01_RESEARCH_DESIGN/`	`study_design.md`	3.7 KB	Study Protocol	**CLEAN** Labels study exploratory pilot; outlines future RCT.
`01_RESEARCH_DESIGN/`	`sample_size_rationale.md`	3.3 KB	Power Analysis	**CLEAN** Mathematical power calculations independently confirmed.
`01_RESEARCH_DESIGN/`	`intervention_protocol.md`	3.7 KB	Lab Protocol	**CLEAN** Standardized time allocations (Phases 1 to 5).
`01_RESEARCH_DESIGN/`	`statistical_analysis_plan.md`	4.6 KB	SAP Trees	**MAJOR** Normality-dependent test selection logic.
`02_ETHICS/`	`ethics_checklist.md`	3.0 KB	IRB Exemption	**BLOCKER** Assumes exemption is self-certifying.
`02_ETHICS/`	`participant_information_sheet.md`	3.1 KB	Participant Info	**CLEAN** Voluntary, anonymous, zero grade consequence.
`02_ETHICS/`	`informed_consent_template.md`	1.7 KB	Consent Form	**CLEAN** Standard opt-in check boxes.
`02_ETHICS/`	`ethics_submission_notes.md`	2.0 KB	IRB Filing Guidance	**CLEAN** 45 CFR 46.104(d)(1) rationale articulated.
`03_ASSESSMENTS/`	`pretest.md`	5.0 KB	Baseline Test	**MAJOR** 70% of answers keyed to option "B".
`03_ASSESSMENTS/`	`posttest.md`	5.4 KB	Parallel Test	**BLOCKER** 80% keyed to "B" (Q1–Q8 all B); instructor cues items.
`03_ASSESSMENTS/`	`answer_key.md`	4.8 KB	Master Keys	**CLEAN** Mechanistic rationales cite John Dolan troubleshooting.
`03_ASSESSMENTS/`	`scoring_rubric.md`	1.8 KB	Practical Rubric	**BLOCKER** 12-pt rubric criteria not supported in CSV data schema!
`03_ASSESSMENTS/`	`practical_diagnostic_task.md`	3.6 KB	Fault Tasks	**CLEAN** Alpha (leak), Beta (void), Gamma (bubble), Delta (contam).

Folder / Module	Artifact Filename	Size	Role	Audit Verification Status
04_USABILITY/	SUS_instrument.md	2.2 KB	SUS Survey	MAJOR Modified Brooke wording in Q1 and Q7.
04_USABILITY/	usability_questionnaire.md	1.7 KB	Targeted UX	CLEAN Kept separate from Brooke SUS calculation.
04_USABILITY/	qualitative_feedback_form.md	3.0 KB	Qualitative UX	CLEAN Inductive thematic coding protocol defined.
05_DATA_TEMPLATES/	anonymized_data_dictionary.md	3.8 KB	Codebook	BLOCKER Lacks variables for practical reasoning and actions.
05_DATA_TEMPLATES/	participant_data_template.csv	592 B	CSV Template	CLEAN 1 single header row, 0 participant records.
05_DATA_TEMPLATES/	data_entry_instructions.md	1.4 KB	Entry Protocol	CLEAN Double-blind entry audit protocol specified.
06_ANALYSIS/	calculate_pre_post_stats.py	3.0 KB	Stats Script	CLEAN Executes cleanly; drops zero-ties in Wilcoxon test.
06_ANALYSIS/	calculate_sus_score.py	2.5 KB	SUS Script	CLEAN Correct odd/even scoring multipliers.
06_ANALYSIS/	calculate_hake_gain.py	1.8 KB	Hake Script	CLEAN Handles ceiling effect edge cases.
06_ANALYSIS/	DEMO_DATA_NOT_REAL.csv	1.9 KB	Synthetic Demo	CLEAN Explicitly labeled; isolated in test_data folder.
07_MANUSCRIPT/	manuscript_outline.md	7.1 KB	JCE Blueprint	CLEAN Full 13-section structure conforming to JCE specs.
07_MANUSCRIPT/	title_options.md	3.5 KB	Title Options	CLEAN Rejects AI, Digital Twin, and universal accuracy.
07_MANUSCRIPT/	provisional_abstract.md	2.5 KB	Placeholder Abs	CLEAN Strictly preserves [N], [MEAN], [SUS] tokens.
07_MANUSCRIPT/	claims_ledger.md	2.8 KB	Claims Register	CLEAN Prohibits digital twins, AI, and mechanistic claims.
07_MANUSCRIPT/	figure_plan.md	3.8 KB	Figure Plan	CLEAN Figures 5–7 held as empty placeholders.
08_REPRODUCIBILITY/	instructor_protocol.md	3.6 KB	Teaching Script	BLOCKER Phase 3 script spoon-feeds exact post-test answers.
08_REPRODUCIBILITY/	reproducibility_checklist.md	1.8 KB	Open Science	CLEAN Open science data/code deposition rules articulated.
08_REPRODUCIBILITY/	study_protocol.md	1.4 KB	Master Protocol	CLEAN Timeline and double-locked paper storage rules.
09_REPORTS/	PHASE4_REPORT.md / .pdf	23 KB	Synthesis Docs	CLEAN Complete synthesis report archived.

4. AUDIT SEVERITY FRAMEWORK

All findings are evaluated against international educational research integrity standards (AERA/APA/NCME Standards for Educational Testing, JCE Author Guidelines, and ICMJE Research Governance):

- `CRITICAL BLOCKER` : A fatal methodological, ethical, or psychometric defect that invalidates data collection, produces unrecoverable data loss, or violates human subject research compliance. **Must be corrected before participant recruitment.**
- `MAJOR ISSUE` : A substantial methodological flaw that compromises statistical power, distorts test validity, or creates severe peer-review vulnerability. **Should be corrected before recruitment unless formally justified.**
- `MINOR ISSUE` : A technical discrepancy, non-critical reporting ambiguity, or documentation inconsistency that does not halt data collection but should be resolved before manuscript submission.
- `OBSERVATION` : A noteworthy implementation characteristic, operational nuance, or neutral feature.
- `STRENGTH` : An exemplary methodological design choice, bias safeguard, or reproducibility feature.

5. SYSTEM USABILITY SCALE (SUS) VALIDITY AUDIT

The System Usability Scale (Brooke 1996) is a standardized, highly calibrated psychometric instrument. Modifying item phrasing breaks standardization and impairs comparison against empirical benchmark distributions (Bangor et al. 2008, Sauro & Lewis 2016).

#	Standard Brooke (1996) Wording	SimuPharm Phase 4 Phrasing	Difference	Methodological Verdict
Q1	"I think that I would like to use this system frequently."	"I think that I would like to use this system frequently *for chromatography learning*."	Added qualifying prepositional phrase.	`MAJOR ISSUE` Narrows construct to course utility, artificially boosting agreement. Restore Brooke wording OR label as "Adapted SUS".
Q2	"I found the system unnecessarily complex."	"I found the system unnecessarily complex."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q3	"I thought the system was easy to use."	"I thought the system was easy to use."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q4	"I think that I would need the support of a technical person to be able to use this system."	"I think that I would need the support of a technical person to be able to use this system."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q5	"I found the various functions in this system were well integrated."	"I found the various functions in this system were well integrated."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q6	"I thought there was too much inconsistency in this system."	"I thought there was too much inconsistency in this system."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q7	"I would imagine that most people would learn to use this system very quickly."	"I would imagine that most *students* would learn to use this system very quickly."	Substituted "people" with "students".	`MINOR ISSUE` Contextual substitution commonly accepted in education, but strictly deviates from canonical Brooke. Declare in Methods.
Q8	"I found the system very cumbersome to use."	"I found the system very cumbersome to use."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q9	"I felt very confident using the system."	"I felt very confident using the system."	Exact Match	`RETAIN` Standard Brooke item preserved.
Q10	"I needed to learn a lot of things before I could get going with this system."	"I needed to learn a lot of things before I could get going with this system."	Exact Match	`RETAIN` Standard Brooke item preserved.

> ⚠️ **SUS Standardization Ruling**
>
> Because Item 1 and Item 7 contain modified wording, the instrument cannot be cited as an unmodified standard Brooke SUS without inviting referee objection. The author must either: (1) Revert Q1 to "*I think that I would like to use this system frequently*" and Q7 to "*most people*" before printing surveys; OR (2) Explicitly define the scale as an **"Adapted Educational System Usability Scale (SUS-Ed)"** and cite Lewis & Sauro (2018) for psychometric equivalence. Domain-specific UX items (`usability_questionnaire.md`) must remain strictly isolated.

6. PRIMARY OUTCOME AND ENDPOINT HIERARCHY

A rigorous educational trial requires a pre-specified, immutable outcome hierarchy to prevent "p-hacking" and selective outcome switching.

PRIMARY OUTCOME (Pre-Specified & Confirmatory)
Paired Troubleshooting Score Gain (ΔScore = Post-Test − Pre-Test, 0 to 10 points)
Inferential Test: Wilcoxon Signed-Rank Test (Two-Tailed, $\alpha = 0.05$). Paired *t*-test pre-specified as sensitivity analysis.

Hierarchy Flow: **PRIMARY ENDPOINT** → **SECONDARY ENDPOINTS (HOLM-BONFERRONI ADJUSTED)** → **EXPLORATORY SUBGROUPS**

SECONDARY OUTCOMES (Multiplicity-Controlled via Holm-Bonferroni Step-Down)
1. **Practical Diagnostic Accuracy:** % Correct across 4 Unknown Simulated Faults (Alpha, Beta, Gamma, Delta)
2. **Diagnostic Latency:** Median Time-to-Diagnosis in seconds (Interquartile Range)
3. **Diagnostic Confidence Calibration:** Pearson correlation *r* between confidence rating (1–5) and diagnostic accuracy
4. **System Usability Scale (SUS):** Mean continuous score (0–100) and Bangor adjective grade
5. **Targeted Pedagogical Usability:** 8 domain-specific UX items (1–5 Likert scale, reported as medians/IQRs)
6. **Hake Normalized Gain:** Average cohort gain ⟨g⟩ = (Post − Pre) / (10 − Pre)
7. **Qualitative Learning Themes:** Inductive thematic coding of open-ended participant reflections

EXPLORATORY SUBGROUP ANALYSES (Hypothesis-Generating, Descriptive Only)
- Disciplinary Background: B.Pharm (n ≈ 12) vs. B.Sc. Chemistry (n ≈ 13)
- Prior HPLC Exposure: Hands-on laboratory instrument experience vs. complete novices
- Item-Level Psychometrics: Item difficulty index (*P*) and discrimination index (*D*)

Hierarchy Audit Ruling: The primary endpoint (ΔScore) is stable across all documents with zero switching detected. However, because 7 secondary endpoints are specified, testing without family-wise error rate control produces a >40% false-positive risk. The SAP must mandate **Holm-Bonferroni correction** for all secondary inferential tests.

7. STATISTICAL ANALYSIS AUDIT & STATISTICAL ISSUE REGISTER

A rigorous audit of the Pre-Specified Statistical Analysis Plan (`statistical_analysis_plan.md`) and analysis code (`calculate_pre_post_stats.py`) identified multiple methodological vulnerabilities:

Table 7.1: Formal Statistical Issue Register

Issue ID	Dimension	Severity	Evidence in Source Documents	Methodological Consequence	Blocker?
STAT-01	Normality-Contingent Test Selection	MAJOR	`statistical_analysis_plan.md` line 28: *"If Shapiro-Wilk $p \geq 0.05$ → Paired t-test; If $p < 0.05$ → Wilcoxon"*	Pre-testing normality in small samples (N=20–30) has low power (<40%) and distorts nominal Type I error rate ($\alpha \neq 0.05$). **Pre-specify Wilcoxon as primary.**	**YES**
STAT-02	Wilcoxon Zero-Difference Handling	MAJOR	`calculate_pre_post_stats.py` line 52: `stats.wilcoxon(post, pre, zero_method='wilcox')`	`zero_method='wilcox'` discards participants with zero score gain ($\Delta = 0$), artificially inflating perceived effect size. **Switch to Pratt's method.**	**YES**
STAT-03	Small-Sample Discrete Ordinality	MAJOR	Pre/post scores are integers from 0 to 10; differences are bounded integers (−10 to +10).	Discrete counts violate continuous Gaussian assumptions. Compute Hodges-Lehmann median difference and bootstrap 95% CIs.	**YES**
STAT-04	Multiple Testing Correction	MINOR	`statistical_analysis_plan.md` mentions Holm-Bonferroni, but analysis script lacks implementation.	Secondary outcomes risk spurious false-positive significance. Integrate `multipletests` with Holm adjustment.	NO
STAT-05	Hake Gain Ceiling Division	MINOR	`calculate_hake_gain.py` line 26: `(post - pre) / (10 - pre)`	Individual gain undefined if Pre = 10. Report cohort average gain $\langle g \rangle$ as primary; exclude undefined cases.	NO
STAT-06	Subgroup Underpowering	MAJOR	`manuscript_outline.md` plans subgroup comparisons (B.Pharm vs B.Sc.; Prior Exp vs Novice).	In N=25, subgroups of n=10–12 have statistical power <25% for moderate effects (d=0.50). Report descriptive stats only.	NO
STAT-07	Time-to-Diagnosis Outlier Skewness	MINOR	Latency measured in seconds (range 10–300 s) is inherently right-skewed.	Mean latency is heavily distorted by distracted students. Mandate **Median Diagnostic Latency (IQR)**.	NO
STAT-08	Missing Data Handling Ambiguity	MINOR	`statistical_analysis_plan.md` does not specify item-level vs participant-level exclusion.	Risk of post-hoc exclusion bias. Pre-specify complete-case analysis for pre/post scores (≥1 missing = excluded).	NO

8. PRACTICAL DIAGNOSTIC TASK AUDIT & SCHEMA RECONCILIATION

A critical structural conflict was discovered between the scoring rubric, assessment task, and data entry schema.

ASSESSMENT INSTRUMENT (practical_diagnostic_task.md)
Students diagnose 4 simulated unknowns and record: (1) Fault Identification, (2) Physical Reasoning, and (3) Corrective Action.

ANALYTICAL SCORING RUBRIC (scoring_rubric.md)
Graders evaluate each scenario on a 3-component rubric: Fault ID (0–2 pts) + Physical Reasoning (0–2 pts) + Corrective Action (0–2 pts) = 0 to 6 points per task (0 to 24 points grand total across 4 tasks) with independent double-rater scoring.

FATAL DATA SCHEMA MISMATCH (participant_data_template.csv)
CRITICAL BLOCKER: The CSV template contains ONLY binary columns: `task_alpha_correct` (0/1) and `task_total_score` (0–4)!
Consequence: The template has NO columns to store physical reasoning scores, corrective action scores, scenario totals, or Rater 1/Rater 2 scores. Data entry clerks will be forced to truncate the entire 12-point rubric into a 4-point binary check, causing irreversible data loss!

To resolve this blocker, the CSV template and data dictionary must be expanded with the following mandatory columns:

Table 8.2: Complete Practical Task Schema Reconciliation

Rubric Dimension	Scoring Scale	Existing CSV Column	Status in Current CSV	Required Mandatory Column Addition
Scenario Alpha: Fault ID	0, 1, 2 points	`task_alpha_correct`	Truncated to 0/1	`task_alpha_id_score` (Integer: 0, 1, 2)
Scenario Alpha: Reasoning	0, 1, 2 points	NONE	**COMPLETELY MISSING**	`task_alpha_reason_score` (Integer: 0, 1, 2)
Scenario Alpha: Action	0, 1, 2 points	NONE	**COMPLETELY MISSING**	`task_alpha_action_score` (Integer: 0, 1, 2)
Scenario Alpha: Total	0 to 6 points	NONE	**COMPLETELY MISSING**	`task_alpha_rubric_total` (Integer: 0 to 6)
Scenario Beta: ID / Reason / Action	0 to 6 points	`task_beta_correct`	Truncated to 0/1	`task_beta_id_score`, `_reason_score`, `_action_score`, `_rubric_total`
Scenario Gamma: ID / Reason / Action	0 to 6 points	`task_gamma_correct`	Truncated to 0/1	`task_gamma_id_score`, `_reason_score`, `_action_score`, `_rubric_total`
Scenario Delta: ID / Reason / Action	0 to 6 points	`task_delta_correct`	Truncated to 0/1	`task_delta_id_score`, `_reason_score`, `_action_score`, `_rubric_total`
Inter-Rater Reliability	Rater 1 vs Rater 2	NONE	**COMPLETELY MISSING**	`rater1_total_score`, `rater2_total_score` (Integers: 0 to 24)
Grand Rubric Total	0 to 24 points	`task_total_score` (0–4)	Severe Scale Mismatch	`task_rubric_grand_total` (replaces 0–4 score)

9. PRE/POST TEST EQUIVALENCE AUDIT & KEYED RESPONSE BIAS

A rigorous comparative item analysis was performed across all 10 pre-test items (`pretest.md`) and 10 post-test items (`posttest.md`).

Pair	Diagnostic Construct	Pre-Test Item Focus	Post-Test Isomorphic Equivalent	Pre Key	Post Key	Equivalence Verdict
Q1	Pump Cavitation	25 bar drop + baseline spikes	Cyclical ±20 bar fluctuation + negative spikes	**B**	**B**	**EXCELLENT** Bloom: Analysis
Q2	Pre-Column Fitting Leak	+30% retention + 85 to 55 bar drop	+25% retention + 90 to 60 bar drop + droplets	**A**	**B**	**EXCELLENT** Bloom: Application
Q3	Column Inlet Voiding	Split peaks + tailing Tf > 2.2	Bifurcated doublet peaks + tailing Tf = 2.4	**B**	**B**	**EXCELLENT** Bloom: Analysis
Q4	Ghost Peaks / Contamination	Unassigned peaks in blank gradient	Ghost peaks doubling with initial hold time	**B**	**B**	**EXCELLENT** Bloom: Comprehension
Q5	High Backpressure Isolation	Sudden overpressure > 400 bar	Disconnecting inlet drops 380 to 5 bar	**B**	**B**	**EXCELLENT** Bloom: Application
Q6	Mobile Phase pH Drift	Selective acidic API drift vs neutral	Ibuprofen shift vs Caffeine/Toluene stability	**B**	**B**	**EXCELLENT** Bloom: Analysis
Q7	Peak Asymmetry Mechanisms	Fronting caused by mass overload	Tailing caused by residual silanol interactions	**B**	**B**	**DISPARITY** Pre: fronting; Post: tailing.
Q8	Cavitation Troubleshooting	First action: open purge valve, prime	Pump degassing sequence: open purge valve	**B**	**B**	**EXCELLENT** Bloom: Application
Q9	Void vs Blockage Diagnosis	Frit blockage vs inlet voiding	Inlet void vs sample solvent mismatch	**A**	**A**	**GOOD** Bloom: Evaluation
Q10	Leak Location Diagnostics	Post-column vs pre-detector leak	Pre-injector leak (area recovery vs retention)	**C**	**A**	**GOOD** Bloom: Evaluation

PSYCHOMETRIC DEFECT: KEYED RESPONSE POSITION BIAS

Inspection of the master answer keys reveals a fatal psychometric flaw:

- **Pre-Test Key Distribution:** Option A = 2, **Option B = 7 (70%)**, Option C = 1, Option D = 0.
- **Post-Test Key Distribution:** Option A = 2, **Option B = 8 (80%)**, Option C = 0, Option D = 0.
- **Critical Flaw: In the post-test, Items Q1 through Q8 are ALL keyed to Option B!** A disengaged student who blindly selects "B" for every item scores 80% on the post-test and 70% on the pre-test. This completely destroys test validity and inflates gain scores. **The answer choices must be re-randomized across A, B, C, and D prior to recruitment.**

10. INSTRUCTOR BIAS AUDIT (TEACHING-TO-THE-TEST FORENSICS)

A line-by-line review of `instructor_protocol.md` (Phase 3: Interactive Fault Exploration, Minutes 18–30) revealed severe instructor demand characteristics and direct test cueing:

BIASED INSTRUCTOR SCRIPT IN SOURCE DOCUMENT (instructor_protocol.md)

"Open the 'Fault Injection' panel:
1. Toggle 'Air Bubble'. Observe the pressure gauge dipping cyclically and sharp baseline spikes. Why does this happen? The air pocket compresses in the pump head, causing momentary flow delivery loss.
2. Toggle 'System Leak'. What happens to retention times? They increase by 30% because liquid is dripping out before the column.
3. Toggle 'Column Void'. Notice the split peaks. Channeling in the collapsed silica bed splits the solute band.
4. Toggle 'Contamination'. Notice the ghost peaks."

FATAL CONFOUND: DIRECT TEST CUEING IDENTIFIED

The instructor script explicitly spoon-feeds the exact symptoms, numerical shifts (-20 bar, -50 bar, +30% tR), and mechanisms tested in **Post-Test Items Q1–Q4** and **Practical Scenarios Alpha–Delta** right before the exam! Post-test gains would measure rote memorization of the instructor's spoken words, NOT simulation learning!

MANDATORY NEUTRAL REPLACEMENT SCRIPT (Non-Directive Guided Inquiry)

Replacement Script: "Open the 'Fault Injection' panel. Activate Fault 1. Examine the pressure gauge and chromatographic canvas carefully. Notice any changes from baseline. Think about what physical change in the fluidic pathway could cause this behavior. Now activate Fault 2, Fault 3, and Fault 4 sequentially. Compare how each fault alters retention, peak shape, and backpressure. Record your observations in your lab notebook."

11. SIMULATION VS PHYSICAL REALITY AUDIT

To maintain scientific documentation integrity, simulated representations must be clearly differentiated from physical chromatography:

Fault Condition	Real-World Physical Phenomenon	SimuPharm Simulated Implementation	Defensible Pedagogical Claim	Prohibited Claim
Air Bubble / Cavitation	Gas bubble in pump check valve dampens displacement stroke, causing stroke-synchronized flow drop and refractive optical spikes.	Discrete `-20 bar` subtraction, `noiseMultiplier: 5`, `rtShift: +0.10` step function.	Qualitatively reproduces characteristic pressure ripple and baseline noise symptomology.	✗ Cannot claim mechanistic two-phase Navier-Stokes cavitation fluid mechanics.
Fitting / System Leak	Mobile phase loss upstream of column reduces superficial linear velocity ($u < u_{set}$), elongating retention proportional to leak split.	Discrete `-50 bar` subtraction, `rtShift: +0.30` (+30% constant stretch).	Demonstrates inverse relationship between volumetric flow delivery and solute retention.	✗ Cannot claim hydrodynamically calibrated leak aperture or mass balance modeling.
Column Inlet Bed Void	Silica bed collapse at inlet creates void headspace; dual velocity channels split migrating solute bands; pressure typically normal or lower.	Discrete `+30 bar` surge, `tailingFactor: 2.50`, `efficiencyLoss: 50%`.	Visualizes split peaks (doublets) and tailing resulting from bed disruption.	✗ Cannot claim porous bed fluid mechanics. (Note: Real voids do not increase pressure unless frit is fouled).
Contamination / Ghost Peaks	Impurities in mobile phase buffers or solvents concentrate on column head and elute during gradient ramps.	Synthetic peaks injected into peak array at fixed retention times with `pressureEffect: +5 bar`.	Demonstrates appearance of non-analyte ghost peaks in quality control blanks.	✗ Cannot claim chemical equilibrium partition modeling of organic solvent trace impurities.

12. CLAIM-BOUNDARY AUDIT

A global text audit was conducted across all Phase 4 documents for prohibited terminology established in Phase 3.5:

Target Term / Concept	Occurrences	Document Locations	Context of Appearance	Audit Verdict
`"Digital Twin"`	5 files	`PHASE4_SCOPE.md`, `claims_ledger.md`, `manuscript_outline.md`	Listed strictly within "PROHIBITED CLAIMS" and prior art disavowals.	**COMPLIANT** Zero affirmative use.
`"Artificial Intelligence" / "AI"`	3 files	`PHASE4_SCOPE.md`, `claims_ledger.md`, `PHASE4_REPORT.md`	Listed strictly within "PROHIBITED CLAIMS".	**COMPLIANT** Zero affirmative use.
`"Machine Learning" / "BKT"`	4 files	`PHASE4_SCOPE.md`, `claims_ledger.md`, `manuscript_outline.md`	Disavowed in Background & Claims Boundary.	**COMPLIANT** Zero affirmative use.
`"2.07%" General Accuracy`	3 files	`PHASE4_SCOPE.md`, `claims_ledger.md`, `PHASE4_REPORT.md`	Cited strictly as an audited in-sample calibration artifact.	**COMPLIANT** Not claimed as general accuracy.
`"Validated Pressure Dynamics"`	2 files	`PHASE4_SCOPE.md`, `claims_ledger.md`	Listed within prohibited claims (identified as theoretical).	**COMPLIANT** Maintained as theoretical.
`"Mechanistic Fault Physics"`	2 files	`PHASE4_SCOPE.md`, `claims_ledger.md`	Listed within prohibited claims (identified as heuristic).	**COMPLIANT** Maintained as heuristic.
`"300+ APIs"`	3 files	`PHASE4_SCOPE.md`, `claims_ledger.md`, `PHASE4_REPORT.md`	Listed within prohibited claims (prototype has 4 APIs).	**COMPLIANT** Maintained 4-analyte scope.
`"Demonstrates" / "Proves"`	3 files	`provisional_abstract.md`, `answer_key.md`, `claims_ledger.md`	Used in abstract: *"findings demonstrate that SimuPharm provides..."*	**MINOR OVERCLAIM** Soften to "suggest" or "indicate".

13. DATA DICTIONARY & MISSING-DATA AUDIT

A cross-check of `anonymized_data_dictionary.md` against questionnaires, rubric, and SAP revealed essential missing variables and pre-specified exclusion rules:

Table 13.1: Data Dictionary Audit & Required Revisions

Category	Audited Parameter	Source Document Conflict	Recommended Resolution
Missing Practical Columns	Reasoning & Action Scores	Present in `scoring_rubric.md`; absent in data dictionary.	Add `task_[x]_reason_score` and `task_[x]_action_score` (Integers 0–2).
Inter-Rater Scoring	Cohen's Kappa (κ)	Required by rubric; no rater columns in dictionary.	Add `rater1_score` and `rater2_score` columns for inter-rater adjudication.
Targeted UX Survey Items	`ux_u1` to `ux_u8`	Data dictionary lists 8 columns; survey has 8 items.	Consistent; retain as secondary ordinal variables.
Hake Gain Denominator	Pre = 10 Division by Zero	`(post - pre) / (10 - pre)` undefined when Pre = 10.	Explicitly code as `NaN` / `N/A` in dictionary and analysis script.
Item-Level Missingness	Incomplete Pre/Post-Tests	Unspecified in current SAP.	**Rule:** If ≥1 question is blank on a 10-item test, total score is coded missing for paired analysis.
SUS Missing Item Rule	Incomplete SUS Questionnaire	Unspecified in current SAP.	**Rule (Sauro 2016):** If 1–2 items missing, substitute neutral score (3); if ≥3 missing, exclude from SUS.
Diagnostic Task Missingness	Skipped Scenario	Unspecified in current SAP.	**Rule:** If a student skips an unknown scenario, score is 0; latency is excluded from median latency.

14. DATA INTEGRITY & NON-FABRICATION VERIFICATION

A physical filesystem audit was conducted to verify that zero synthetic data has contaminated the participant template:

- **Blank Template Verification:** `participant_data_template.csv` contains exactly **1 single header row (592 bytes) and 0 participant rows**. Zero fabricated data confirmed.
- **Demo Data Isolation:** The 5 synthetic test records used to verify analysis scripts are sequestered in `06_ANALYSIS/test_data/DEMO_DATA_NOT_REAL.csv` and explicitly labeled. Analysis code cannot accidentally read this file during live study execution.
- **De-Identification Safeguards:** Participant IDs are randomized (`P01` to `P25`). Physical paper consent forms are stored separately from assessment sheets in a double-locked cabinet.

15. ETHICS READINESS AUDIT (EXEMPTION VS AUTHORIZATION)

A critical distinction exists in human subjects research between *qualifying for an ethics exemption* and *possessing institutional authorization to recruit participants*.

STEP 1: PROTOCOL SELF-ASSESSMENT (Completed in Phase 4)
Author drafted 45 CFR 46.104(d)(1) exemption rationale and prepared participant information/consent forms.

Ethics Progression: **STEP 1: SELF-ASSESSMENT** → **STEP 2: FORMAL FILING (MANDATORY NEXT STEP)** → **STEP 3: WRITTEN LETTER**

CRITICAL BLOCKER: INVESTIGATORS CANNOT SELF-EXEMPT RESEARCH
In `ethics_checklist.md`, Exemption Category 1 is checked as "COMPLIANT". However, under academic research policies and US federal regulations (OHRP), **investigators cannot self-exempt their own research involving students**. An official Institutional Review Board (IRB) or Departmental Research Ethics Committee must issue a formal written exemption letter.
Recruitment Status: STRICTLY NO-GO until a written institutional exemption letter or departmental authorization memo is signed.

16. SAMPLE-SIZE AUDIT & POWER VERIFICATION

Independent mathematical verification of the sample-size calculations in `sample_size_rationale.md` was conducted via standard non-central *t*-distribution formulations:

Sample Size (N)	Effect Size (d_z)	Reported Power	Recomputed Power	SUS Margin of Error	Methodological Assessment
N = 15	d_z = 0.70 (Large)	71.0%	**71.2%**	±5.82 points	Minimum feasible pilot; underpowered for moderate effects (d=0.50, power 46%).
N = 22	d_z = 0.65 (Mod-Large)	85.0%	**85.1%**	±4.66 points	Exact mathematical sample size requirement for 85% power at α = 0.05.
N = 25	**d_z = 0.65**	**89.2%**	**89.3%**	**±4.33 points**	**Preferred Pilot Target:** Excellent power; manageable single-section cohort.
N = 25	d_z = 0.80 (Large)	97.5%	**97.6%**	±4.33 points	Near-certain statistical detection if software produces large effect.
N = 40	d_z = 0.50 (Moderate)	97.8%	**87.1%**	±3.32 points	Discrepancy: Report claimed 97.8% for d=0.50; true power for d=0.50 is 87.1%.

> **Power Analysis Verdict**
>
> The sample size target of **N = 20–30 participants** is mathematically sound and realistically achievable in a single university laboratory section. It provides >85% power to detect a moderate-to-large paired diagnostic gain ($d_z \geq 0.65$) and estimates the mean SUS score within ±4.3 points. However, the author must explicitly emphasize in the manuscript that N=25 establishes **pilot feasibility and local instructional utility**, not population-level generalizability.

17. ANALYSIS PIPELINE CODE VERIFICATION

All pre-specified Python analysis scripts in `06_ANALYSIS/analysis_code/` were executed against the labeled demonstration dataset (`DEMO_DATA_NOT_REAL.csv`):

```
=== EXECUTION AUDIT LOG: calculate_pre_post_stats.py ===
Data source: 06_ANALYSIS/test_data/DEMO_DATA_NOT_REAL.csv
Sample Size (N): 5 | Pre-Test Mean: 4.00 (SD: 1.58) | Post-Test Mean: 7.80 (SD: 1.30)
Mean Paired Gain: +3.80 points (SD: 0.84) | Median Gain: +4.0
Shapiro-Wilk Test: W = 0.8810, p = 0.3140 (Normal distribution not rejected)
Paired t-test: t(4) = 10.156, p = 5.29e-04, 95% CI: [+2.76, +4.84], Cohen's dz = 4.54
Wilcoxon Signed-Rank: W = 0.0, p = 0.0625, Rank-Biserial r = 1.00
STATUS: PASS (Execution verified; Pratt zero-difference method recommended)

=== EXECUTION AUDIT LOG: calculate_sus_score.py ===
Usable SUS Submissions (N): 5 | Mean SUS: 81.00 (SD: 14.43) | Median: 75.00 (IQR: 22.50)
95% Confidence Interval: [63.09, 98.91] | Bangor Adjective Rating: Excellent
STATUS: PASS (Brooke 1996 multipliers verified)

=== EXECUTION AUDIT LOG: calculate_hake_gain.py ===
Overall Average Normalized Gain <g>: 0.633 | Mean Individual Gain: 0.654 (SD: 0.143)
Hake Classification: Medium Gain (0.30 ≤ g < 0.70)
STATUS: PASS (Ceiling condition handled)
```

18. FIGURE & MANUSCRIPT CONSISTENCY AUDIT

Cross-checking the 7 planned figures (`figure_plan.md`) against the data schema and manuscript outline revealed complete alignment:

- **Figure 1 (UI Screenshots):** Captures actual frozen SimuPharm frontend controls (sliders, canvas, fault toggles).
- **Figure 2 (Simulation Workflow):** Accurately illustrates data flow without claiming AI or digital twin loops.
- **Figure 3 (Normal Baseline):** Displays standard analgesic mixture (50% B, 1.0 mL/min, 30°C, C18).
- **Figure 4 (Fault Signatures):** Four-panel layout depicting air bubble, leak, void, and contamination.
- **Figures 5, 6, 7 (Student Data):** Correctly designated as **unpopulated placeholders** until real data collection. Figure 6 (unknown task accuracy/latency) can be generated once the schema reconciliation table is merged into the CSV template.

19. MANUSCRIPT LANGUAGE AUDIT (SENSITIVE VERB SCRUBBING)

All manuscript sections were audited for overreaching verbs and claims:

Sensitive Phrase	Context in Manuscript Draft	Auditor Assessment	Mandatory Replacement Wording
`"demonstrates"`	Abstract line 18: *"findings demonstrate that SimuPharm provides..."*	Too definitive for an exploratory pilot study without a control group.	Replace with **"suggest"**, **"indicate"**, or **"provide preliminary evidence that"**.
`"proves"`	Answer Key Q5: *"proves the hydraulic restriction is inside the column..."*	Acceptable in diagnostic deduction, but dangerous if applied to software efficacy.	In pedagogical context: replace with **"indicates"** or **"isolates"**.
`"validated"`	Folder name & report titles: *"Educational Validation Study"*	Overreaching; implies formal psychometric construct validation.	Replace with **"Educational Evaluation"** or **"Pilot Usability Assessment"**.
`"improves"`	Claims ledger: *"SimuPharm improves HPLC troubleshooting skills"*	Causal claim; cannot be definitively established without a control group.	Replace with **"is associated with measurable short-term score gains"**.
`"superior"`	Claims ledger: *"superior to traditional textbook manuals"*	Correctly placed in Tier 2 (prohibited for present pilot).	Retain in prohibited tier.
`"transfer"`	Manuscript outline Section 10: *"Transfer to physical instruments remains unmeasured"*	Properly framed as a candid limitation.	Retain as an explicit limitation.

20. CROSS-DOCUMENT CONSISTENCY MATRIX

A master cross-document consistency audit was performed across all 18 key planning artifacts:

Document A	Document B	Target Dimension	Consistent?	Identified Conflict & Mandatory Resolution
`research_questions.md`	`statistical_analysis_plan.md`	Primary Outcome	YES	Both define ΔScore = Post − Pre on 0–10 scale. Harmonized.
`sample_size_rationale.md`	`study_design.md`	Cohort Size	YES	Both specify N = 20–30 undergraduate students. Harmonized.
`scoring_rubric.md`	`participant_data_template.csv`	Practical Schema	NO	Rubric scores 12 points (reasoning/action); CSV only has 0/1 binary! **Expand CSV template with Table 8.2 columns.**
`instructor_protocol.md`	`posttest.md`	Instructor Cueing	NO	Instructor script spoon-feeds exact post-test answers during Phase 3! **Replace script with neutral guided inquiry.**
`pretest.md`	`posttest.md`	Keyed Keys	NO	70% of pre-test and 80% of post-test are keyed to "B" (Q1–Q8 all B!). **Re-randomize answer options across A, B, C, D.**
`SUS_instrument.md`	Brooke (1996) Canonical	Item Phrasing	NO	Modified wording in Items 1 and 7 ("for chromatography learning", "students"). **Restore canonical wording or label "Adapted SUS".**
`statistical_analysis_plan.md`	`calculate_pre_post_stats.py`	Wilcoxon Ties	NO	Script drops zero-differences (`zero_method='wilcox'`); SAP implies full cohort. **Change script to Pratt's method.**
`ethics_checklist.md`	`PHASE4_DECISION_STATUS.md`	Approval Status	NO	Checklist claims "COMPLIANT"; decision status has approval box unchecked. **Clarify: exemption rationale drafted; institutional sign-off pending.**
`manuscript_outline.md`	`figure_plan.md`	Figure Mapping	YES	All 7 figures match manuscript section references. Harmonized.
`claims_ledger.md`	`PHASE4_SCOPE.md`	Claim Boundary	YES	Identical 3-tier boundary preserved. Harmonized.

21. TOP 10 THINGS THE AUTHOR MUST FIX BEFORE RECRUITMENT

The following ranked register specifies the exact mandatory corrections that must be executed prior to recruiting human student participants:

Rank	Issue Category	The Exact Problem	Why It Matters	Mandatory Correction	Severity
1	Data Schema Mismatch	`scoring_rubric.md` evaluates practical tasks on a 12-pt rubric, but `participant_data_template.csv` only provides binary 0/1 columns.	**Fatal Data Loss:** Reasoning and action scores cannot be stored. Grader inter-rater reliability (κ) cannot be calculated.	Expand CSV template and data dictionary with Table 8.2 columns (`_reason_score`, `_action_score`, `rater1`, `rater2`).	BLOCKER
2	Instructor Test Cueing	`instructor_protocol.md` Phase 3 explicitly recites the exact symptoms, pressure drops, and mechanisms to the class right before testing.	**Fatal Demand Bias:** Post-test gains merely measure short-term auditory recall of instructor statements rather than simulation learning.	Rewrite Phase 3 script to use neutral guided inquiry (instruct students to observe canvas and hypothesize without answers).	BLOCKER
3	Keyed Response Imbalance	70% of pre-test and 80% of post-test items are keyed to Option "B" (Post-test Items Q1 through Q8 are ALL B).	**Psychometric Invalidation:** A student guessing "B" blindly scores 80% on post-test. Destroys test validity and inflates gains.	Re-scramble answer choices across A, B, C, and D so that each option represents ≈25% of keys.	BLOCKER
4	Ethics Authorization Status	`ethics_checklist.md` declares 45 CFR 46 exemption "COMPLIANT", but no institutional IRB/faculty approval letter has been granted.	**Regulatory Non-Compliance:** Investigators cannot self-exempt research involving human students. Violates publication ethics.	Submit protocol to institutional IRB or Department Chair; obtain written formal determination letter before recruiting.	BLOCKER
5	SUS Canonical Wording	Items 1 and 7 in `SUS_instrument.md` add "for chromatography learning" and substitute "students" for "people".	**Benchmark Incomparability:** Modified wording prevents legitimate comparison against Brooke (1996) normative databases.	Restore exact Brooke canonical wording OR explicitly designate as "Adapted Educational SUS" in all documents.	MAJOR
6	Normality-Contingent SAP	SAP conditions test selection on Shapiro-Wilk testing ($p \geq 0.05 \rightarrow$ t-test; $p < 0.05 \rightarrow$ Wilcoxon).	**Type I Error Distortion:** Normality testing in N=25 has low power; conditioning test choice alters nominal $\alpha = 0.05$ level.	Pre-specify Wilcoxon Signed-Rank Test as primary (due to discrete count data) and paired t-test as sensitivity analysis.	MAJOR
7	Wilcoxon Zero-Ties	`calculate_pre_post_stats.py` uses `zero_method='wilcox'`, which discards students with zero score gain ($\Delta = 0$).	**Artificial Effect Inflation:** Dropping non-improving students artificially inflates the rank-sum statistic and effect size.	Modify script to use `zero_method='pratt'`, which includes tied zero differences in the ranking.	MAJOR
8	Missing Data Rules	No formal pre-specified policy for incomplete assessment sheets or missing SUS items.	**Post-Hoc Data Manipulation:** Graders might arbitrarily discard low-scoring incomplete sheets.	Add explicit missing-data rules to SAP: complete-case analysis for pre/post scores; Sauro imputation for ≤2 missing SUS items.	MAJOR
9	Secondary Multiplicity	Fifteen secondary endpoints lack formal family-wise error rate control.	**False Discovery Inflation:** Testing multiple endpoints at $\alpha = 0.05$ creates >40% risk of spurious significance.	Integrate Holm-Bonferroni step-down correction into the secondary outcome analysis pipeline.	MINOR
10	Abstract Tone Calibration	Abstract states: "*findings demonstrate that SimuPharm provides...*"	**Overreaching Language:** Single-group pilot cannot "demonstrate" general instructional efficacy.	Soften abstract phrasing to: "*preliminary pilot findings suggest that SimuPharm provides...*"	MINOR

22. FROZEN ITEMS (THINGS THAT MUST NOT BE CHANGED)

While the educational protocol and data templates must be corrected, the following scientific boundaries are **STRICTLY IMMUTABLE**:

1. **Frozen SimuPharm Engine:** Zero modifications to `hplc_simulator_script.js` or `script.js`. Do not "fix" the hardcoded dead time (t_0 = 0.50 min), uncalibrated S slopes, or heuristic fault parameters. The educational study must evaluate the software exactly as it exists.
2. **Non-Fabrication Rule:** Zero generation of fake student scores, simulated p-values, artificial SUS ratings, or synthetic qualitative quotes. `participant_data_template.csv` must remain 100% empty until real students participate.
3. **Phase 3.5 Claim Boundaries:** Strict ban on "Digital Twin", "Artificial Intelligence", "Machine Learning", "Bayesian Knowledge Tracing", "universal predictive accuracy", and "experimentally validated pressure/fault physics".
4. **Physical HPLC Separation:** No physical HPLC experiments shall be initiated during Phase 4; JCE Technology Reports evaluate educational utility, not laboratory sensor validation.

23. FINAL RECRUITMENT READINESS DECISION

AUDIT EVALUATION CLASSIFICATION:

NO-GO — MAJOR METHODOLOGICAL CORRECTIONS REQUIRED

JUSTIFICATION & AUDIT SYNTHESIS:

Participant recruitment must not proceed under the current package. While the software architecture is robust, the educational pilot protocol suffers from four fatal defects that would invalidate the research:

1. Data Entry Schema Truncation (Critical Blocker): The data entry template (`participant_data_template.csv`) is missing columns to store the 12-point practical diagnostic task rubric scores (reasoning, corrective action, and inter-rater reliability), which would cause permanent data loss during collection.
2. Instructor Teaching-to-the-Test Bias (Critical Blocker): The instructor protocol in Phase 3 explicitly spoon-feeds the exact fault symptoms, numerical shifts (-20 bar, -50 bar), and mechanisms to the students right before testing, confounding post-test score gains with short-term auditory recall.
3. Extreme Keyed Response Position Bias (Critical Blocker): The pre-test and post-test instruments suffer from severe response position bias, with 70% and 80% of answers keyed to Option "B" (Post-test Items Q1 through Q8 are all B), allowing blind guessing to score 80%.
4. Unverified Institutional Ethics Exemption (Critical Blocker): Institutional ethics exemption has been self-declared but not formally confirmed by an institutional IRB or faculty review body. Under institutional research governance, investigators cannot self-exempt research involving human students.

Once these four critical blockers are corrected—by reconciling the CSV schema, neutralizing the instructor script, re-scrambling the assessment keys across A/B/C/D, and securing written institutional authorization—the study will achieve full methodological rigor and will be approved for recruitment.

CRITICAL PATH ACTIONS BEFORE GO:

1. Update `participant_data_template.csv` and `anonymized_data_dictionary.md` with Table 8.2 columns.
2. Neutralize `instructor_protocol.md` Phase 3 to eliminate direct answers and mechanisms.
3. Re-randomize answer keys in `pretest.md` and `posttest.md` across options A, B, C, D.
4. Submit protocol to institutional ethics authority and secure written exemption letter.
5. Update `calculate_pre_post_stats.py` to use `zero_method='pratt'` for Wilcoxon signed-rank test.

Hostile Pre-Pilot Scientific Audit Panel • Completed & Certified October 4, 2026 • SimuPharm.io Publication Due Diligence